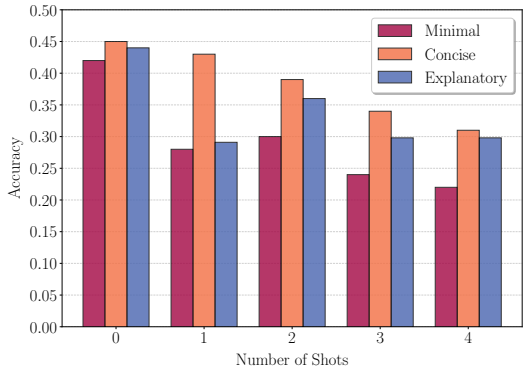


Figure 5: Few-shot prompting accuracy on GPQA-Diamond* dataset using reasoning Qwen-7B (distilled from DeepSeek-R1). The accuracy drops dramatically with respect to the increasing number of examples within the prompt.

Prompting Style Modulates RLM Performance. Our results reveal a consistent and striking trend: increasing the number of in-context examples leads to a monotonic decline in accuracy, regardless of demonstration style (Figure 5). However, the severity of this decline is strongly related to the structure and verbosity of the provided exemplars, with `Minial` being the most RLM-unfriendly prompting style.

While there is a hypothesis explaining that few-shot prompting leads to a reduction in total length of I/O tokens (Figure 6), raw length alone does not fully explain the loss of reasoning effectiveness. Instead, these phenomena call for deeper understanding and explanations for the observed degradation in model performance.

Structural Shifts Triggered by Prompting Styles To better understand these performance variations, we examine the corresponding changes in reasoning graph topology across prompting conditions (Figure 8). It turns out that zero-shot prompting induces richer, more complex graph structures: graphs feature higher exploration density, greater branching and convergence, and a more diverse distribution of reasoning step counts (Figure 7). This suggests that, when not being influenced by extra demonstrations, the model engages in more adaptive and active exploration, revisiting, branching, and synthesizing reasoning steps.

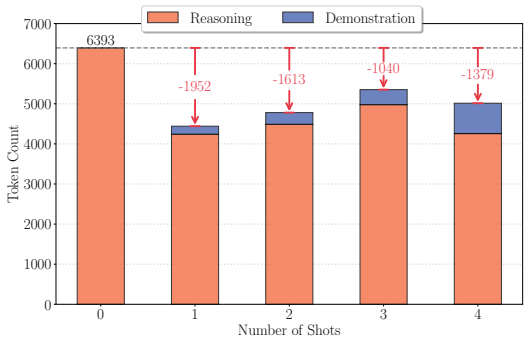


Figure 6: Average number of tokens under different numbers of shots with `explanatary` few-shot style. Few-shot prompting leads to significantly fewer reasoning tokens compared with zero-shot prompting.

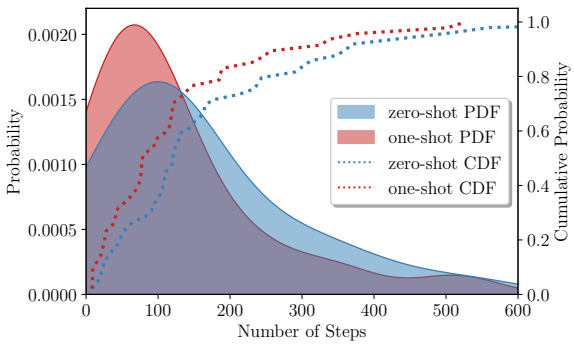
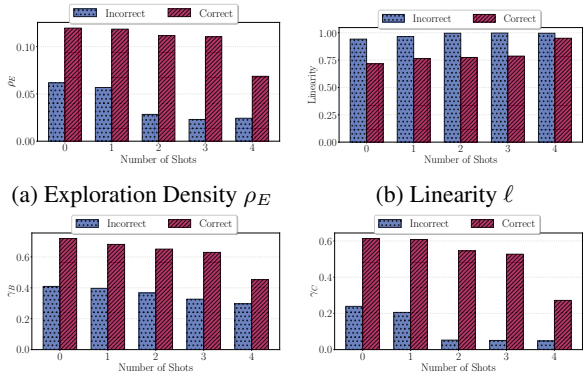


Figure 7: Distribution of reasoning step counts under zero-shot and one-shot prompting. The inclusion of a single demonstration in one-shot settings causes a pronounced shift in the distribution compared to zero-shot, highlighting the sensitivity of RLM to prompt design.



(a) Exploration Density ρ_E (b) Linearity ℓ

(c) Branching Ratio γ_B (d) Convergence Ratio γ_C

Figure 8: Different metrics of reasoning graph given different numbers of few-shot examples.

In contrast, increasing the number of few-shot exemplars, especially in more verbose forms, systematically reduces both branching and convergence ratios, resulting in more linear graph architectures. The model appears to mimic the structure of the provided examples, limiting its capacity for active online reasoning. Notably, even a single demonstration can trigger a significant distributional shift toward shorter, more stereotyped

Prompt Type	Acc (%)	ρ_E	γ_B	γ_C	ℓ	Mean Steps
LLAMA-8B*						
Zero-shot	44.5	0.117	0.564	0.676	0.744	11.6
Concise	41.2	0.065	0.252	0.426	0.947	8.6
Explanatory	32.7	0.057	0.238	0.392	0.926	10.8
QWEN-14B*						
Zero-shot	51.8	0.122	0.612	0.719	0.716	15.8
Concise	48.5	0.069	0.264	0.453	0.931	12.2
Explanatory	45.3	0.061	0.243	0.420	0.946	14.2
QWEN3-32B						
Zero-shot	56.2	0.188	0.835	0.760	0.444	19.0
Concise	53.7	0.110	0.529	0.634	0.766	15.6
Explanatory	50.1	0.069	0.283	0.449	0.931	18.0

Table 1: Comparison of model performance and reasoning graph metrics across different model sizes and prompting paradigms. * denotes reasoning models that are officially distilled from DEEPSEEK-R1.

reasoning chains.

These findings underscore the critical role of few-shot prompting styles in shaping RLM reasoning. Including no or a few extra demonstrations encourages flexible exploration and integration, which fosters more RLM's innate sophisticated reasoning graphs and leads to higher task performance. This motivates a shift in prompt engineering: effective demonstrations should balance informativeness with structural diversity, avoiding excessive *unnecessary* context that suppresses the model's reasoning potential.

4.3 Structural Signatures of Effective Reasoning

To comprehensively reveal the relationship between reasoning structure and task performance, we present in Table 1 a systematic comparison of graph-based metrics across multiple model scales and prompting paradigms. Several key patterns emerge that robustly distinguish effective reasoning.

Sophisticated Reasoning Graph Structure Drive Success. Across all models and prompt types, higher accuracy is consistently associated with richer reasoning graph structure: increased exploration density (ρ_E), higher branching ratio (γ_B), and greater convergence ratio (γ_C). Notably, larger models (e.g., Qwen3-32B) exhibit both the highest accuracy and the most complex graph structures, particularly under zero-shot prompting. This indicates that effective reasoning is achieved through a harder exploration (multiple attempts generation) and integrative convergence (synthesizing reasoning threads).

Prompt Constraints Induce Linearity and Impair Performance. Prompt types that impose stronger structural constraints consistently yield lower branching and convergence ratios, along with increased linearity (ℓ). This shift toward more linear graph topologies is directly correlated with performance degradation. The effect is most pronounced in smaller models but persists even for larger architectures. These results again highlight the double-edged nature of few-shot demonstrations for RLM.

Quantitative Correlations. We extend the Pearson correlation analysis to all four graph-based metrics and observe that exploration density ($r = 0.68$), branching ratio ($r = 0.67$), and convergence ratio ($r = 0.68$) each exhibit a strong positive association with accuracy with all significant at the 0.05 level. These results indicate that denser, more exploratory and convergent reasoning paths are closely linked to model accuracy. Notably, these trends persist across all model scales and prompting regimes, highlighting the robustness and explanatory value of our structural framework.

In summary, these results provide compelling evidence demonstrating that reasoning graph analysis provides highly correlated and deep insights into the internal cognitive dynamics of reasoning language models.

5 Conclusion

This paper introduces a novel graph-based framework for analyzing reasoning output produced by reasoning large language models, offering quantifiable findings about how models *organize* their thought flow under various factors. We first propose a reasoning graph toolkit that efficiently converts raw Chain-of-Thought tokens into analyzable graph structures. We then offer discovery regarding how various prompting styles may cast significant influence on RLM's internal reasoning structure and thus final performance. We also provide strong evidence supporting that graph-level predictors strongly correlate with RLM problem-solving performance. These findings not only establish quantitative insights for future prompt engineering for reasoning models, but also provide a new structural perspective for evaluating reasoning quality beyond traditional metrics and interpreting how LLMs reason at a higher level.

6 Limitations & Future Work

While our current graph-based analysis of RLMs focuses primarily on mathematical and coding tasks, extending this framework to broader domains, such as multi-modal or open-domain reasoning, may yield deeper insights into model behavior across varied scenarios and further inform our understanding of test-time scaling. In addition, the quantifiable structural metrics we propose provide a foundation for future research to explore more localized patterns and relational dynamics within reasoning graphs. We believe that ongoing work along these lines can contribute to a more comprehensive understanding and interpretability of large language models in practice.

References

Anthropic. 2025. Claude's extended thinking. https://www.anthropic.com/news/visible-extended-thinking. Accessed: 2025-05-18.

Marthe Ballon, Andres Algaba, and Vincent Ginis. 2025. The relationship between reasoning and performance in large language models – o3 (mini) thinks harder, not longer. *ArXiv*.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language models are few-shot learners. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS '20, Red Hook, NY, USA. Curran Associates Inc.

Qiguang Chen, Libo Qin, Jinhao Liu, Dengyun Peng, Jiannan Guan, Peng Wang, Mengkang Hu, Yuhang Zhou, Te Gao, and Wanxiang Che. 2025. Towards reasoning era: A survey of long chain-of-thought for reasoning large language models. https://arxiv.org/abs/2503.09567. ArXiv preprint arXiv:2503.09567.

Xingyu Chen, Jiahao Xu, Tian Liang, Zhiwei He, Jianhui Pang, Dian Yu, Linfeng Song, Qiuzhi Liu, Mengfei Zhou, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, Haitao Mi, and Dong Yu. 2024. Do not think that much for 2+3=? on the overthinking of o1-like llms. https://arxiv.org/abs/2412.21187. ArXiv preprint arXiv:2412.21187.

Sergio Da Silva. 2023. System 1 vs. system 2 thinking. *Psych*, 5(4):1057–1076.

DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z.F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, et al. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*.

DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, Damai Dai, Daya Guo, Dejian Yang, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Han Bao, Hanwei Xu, Haocheng Wang, Haowei Zhang, Honghui Ding, Huajian Xin, Huazuo Gao, Hui Li, Hui Qu, J. L. Cai, Jian Liang, Jianzhong Guo, Jiaqi Ni, Jiashi Li, Jiawei Wang, Jin Chen, Jingchang Chen, Jingyang Yuan, Junjie Qiu, Junlong Li, Junxiao Song, Kai Dong, Kai Hu, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Lei Xu, Leyi Xia, Liang Zhao, Litong Wang, Liyue Zhang, Meng Li, Miaojun Wang, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingming Li, Ning Tian, Panpan Huang, Peiyi Wang, Peng Zhang, Qiancheng Wang, Qihao Zhu, Qinyu Chen, Qiushi Du, R. J. Chen, R. L. Jin, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, Runxin Xu, Ruoyu Zhang, Ruyi Chen, S. S. Li, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shaoqing Wu, Shengfeng Ye, Shirong Ma, Shiyu Wang, Shuang Zhou, Shuiping Yu, Shunfeng Zhou, Shuting Pan, T. Wang, Tao Yun, Tian Pei, Tianyu Sun, W. L. Xiao, Wangding Zeng, Wanjia Zhao, Wei An, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, X. Q. Li, Xiangyue Jin, Xianzu Wang, Xiao Bi, Xiaodong Liu, Xiaohan Wang, Xiaojin Shen, Xiaokang Chen, Xiaokang Zhang, Xiaosha Chen, Xiaotao Nie, Xiaowen Sun, Xiaoxiang Wang, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xingkai Yu, Xinnan Song, Xinxia Shan, Xinyi Zhou, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, Y. K. Li, Y. Q. Wang, Y. X. Wei, Y. X. Zhu, Yang Zhang, Yanhong Xu, Yanping Huang, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Li, Yaohui Wang, Yi Yu, Yi Zheng, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Ying Tang, Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yu Wu, Yuan Ou, Yuchen Zhu, Yuduan Wang, Yue Gong, Yuheng Zou, Yujia He, Yukun Zha, Yunfan Xiong, Yunxian Ma, Yuting Yan, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Z. F. Wu, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhen Huang, Zhen Zhang, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhibin Gou, Zhicheng Ma, Zhigang Yan, Zhihong Shao, Zhipeng Xu, Zhiyu Wu, Zhongyu Zhang, Zhuoshu Li, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Ziyi Gao, and Zizheng Pan. 2024. Deepseek-v3 techni-